

The Next Step to Mastering ns-3

In this, the second article on ns-3, the author touches upon Doxygen, Waf, socket programming, etc.



ns-3

Let us continue our journey to master ns-3. We are not yet ready to foray into its strongly fortified castle so, instead, let's explore Doxygen, Waf and socket programming. Once they are conquered, we will be ready for a direct attack on ns-3.

Documenting ns-3 with Doxygen

Programming 101 teaches us that comments make programs more understandable. We also learn that comments are removed by the compiler at the first phase itself. But hats off to the ingenuity of Dimitri van Heesch who created Doxygen, which made the whole business of commenting a lot more exciting. Doxygen is a document preparation tool used by software developers to automatically generate documentation for projects, provided they are commented with tags known to Doxygen. To a developer, this is a real gift but to a novice it might be a slight hindrance. Unlike normal comments, in Doxygen, comments are not intended to help the person who reads the source file directly; rather, they aid in the automatic document generation. Initially, when you go through the source files of ns-3, which are plagued with Doxygen style comments, you might find them unattractive. Figure 1 shows an ns-3 source file with Doxygen comments.

So is it necessary to learn Doxygen to master ns-3? The short answer is 'No'. Let me elaborate—if you know Doxygen, well, that's great! You will have a clear idea where to look for answers in the ns-3 documentation. But if you are not familiar with Doxygen, then all you have to do is remember the fact that whatever text comes in between `/**` and `*/` in an ns-3 source file is a Doxygen comment, and it in no way affects the working of the ns-3 source files written in C++.

All commands start with a backslash (\) or an at-sign (@). For example, you might see commands like `\brief`, `\author`, `\param`, `\bug`, etc, inside Doxygen style comments. They convey a special message to the Doxygen parser. The command `\brief` is used to start a paragraph that serves as a brief description. `\author` starts a new paragraph to display the author names. `\param` starts a parameter description for a function parameter with a name, followed by a description of the parameter. `\bug` is used for bug reporting. There are many more commands used in Doxygen, but you need to look them up only if a need arises. So for now, let us not worry much about Doxygen but move forward and tackle other issues relevant to ns-3.

Installing ns-3 with Waf

Software installation on Linux is often considered troublesome by new users and gives them nightmares. To simplify the installation procedure, ns-3 uses a Python based build automation tool called Waf. Those who are familiar with ns-2 might remember that a tool called *make* was the build automation tool in it. But in ns-3, *make* is replaced by Waf; a relatively modern tool.

ns-3 can be installed by using individual source files or a tarball (archive) release of the source files. I will only discuss the installation of ns-3.22 (the latest version of ns-3) with the tarball release of source files. But remember, with different operating systems and different versions, ns-3 installation might fail occasionally. If that happens, you are the victim of a problem called *dependency hell*. In case of a failed installation, please follow the detailed installation steps given at www.nsnam.org/docs/release/3.22/tutorial/html/getting-started.html#building-ns3. To follow the installation steps of ns-3.22 given here, you need to download a file named `ns-allinone-3.22.tar.bz2`. Please remember that this file alone will enable you to install ns-3 with Waf. The file can be downloaded from www.nsnam.org/release/. It is an archived and compressed file with a size about 25MB. The extension `.tar` denotes an archive file and the extension `.bz2` is associated with the file compression utility `bzip2`. The installation requires GCC tools like `g++`. But usually, the Linux environment will contain all the required tools, by default. So the only thing to ensure is that you have Waf installed in your system prior to ns-3 installation.